From Potentials and Chemicals to the Mind

The Neurological and Psychological Foundations of the Mind

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Contents

1 Nervous System Structure

Before we can talk about thoughts, feelings, memories, or the mind itself, we need a map. The nervous system is the organ that does all of this, it senses, decides, remembers, imagines, and like any complicated machine it helps enormously to know, at least roughly, where its parts are and what each part is for.

At the highest level the system splits into two halves. The **central nervous system (CNS)** is the brain and the spinal cord: the headquarters, where information is integrated and decisions are made. The **peripheral nervous system (PNS)** is everything else, the sprawling network of nerves that links the CNS to muscles, glands, and sensory organs. Every sensation you have ever had and every voluntary movement you have ever made has crossed the boundary between these two systems.

In the rest of this section we walk through the major structures of the CNS and PNS from the top down. By the end you should be able to point, more or less, to where anything we discuss later actually lives.

1.1 The Cerebrum

The cerebrum is the part of the brain you can picture without effort, the wrinkled, walnut-shaped dome that sits on top of everything else. It is also, by some distance, the largest part of the brain and the part most closely associated with what we usually mean by "the mind." Thinking, reasoning, memory, language, voluntary movement, the experience of being a self, all of these depend on cerebral machinery.

Structurally it is symmetrical. Two hemispheres, left and right, are joined by a thick band of white matter called the **corpus callosum**, which lets them share information. Each hemisphere is then divided into four lobes, and each lobe has a characteristic job:

- **Frontal lobe**, voluntary movement, planning, decision-making, and much of what we call personality.
- **Parietal lobe**, processing of bodily sensations such as touch, temperature, and pain, and spatial awareness.
- **Temporal lobe**, hearing, language comprehension, and the formation of long-term memories.
- **Occipital lobe**, vision.

The outer layer of the cerebrum, the **cerebral cortex**, is only a few millimetres thick but is crumpled into deep folds. The folding is not cosmetic: by crumpling the sheet, evolution has packed far more cortical area into the skull than would otherwise fit, and cortical area turns out to matter enormously for cognitive capacity.

1.2 The Cerebellum

Tucked beneath the back of the cerebrum, the cerebellum ("little brain") looks almost like a miniature version of its larger sibling. Despite its size, it contains roughly half of all the neurons in the brain, and it is indispensable for smooth, coordinated movement.

Crucially, the cerebellum does not initiate actions. Instead it receives a copy of the motor commands being issued by the cortex together with sensory feedback about what the body is actually doing, and it compares the two. When the intention and the reality diverge, the cerebellum quietly corrects the discrepancy in real time. That is why skilled movements, reaching for a mug, playing a scale on the piano, walking in a straight line, feel effortless.

When the cerebellum is damaged, this silent correction falls apart. The classic picture includes *ataxia* (loss of coordination), tremor, and difficulty maintaining balance. Patients often describe it as feeling perpetually drunk, which is itself a hint: alcohol disrupts cerebellar function, and that is exactly why it impairs coordination.

1.3 The Midbrain and Brainstem

If the cerebrum is where you think and the cerebellum is where you coordinate, the brainstem is where you stay alive. It connects the brain to the spinal cord and is the conduit for almost every signal passing between them. It is also the home of the circuits that keep you breathing, keep your heart beating, and keep you conscious.

It is traditionally divided into three levels, each with distinct responsibilities:

- **Midbrain**, visual and auditory reflexes, aspects of motor control, and arousal.
- **Pons**, a relay between the cerebrum and cerebellum, and a participant in sleep and respiration.
- **Medulla oblongata**, the control centre for vital autonomic functions: heart rate, blood pressure, breathing, coughing, swallowing.

Because the brainstem handles these non-negotiable tasks, even small lesions there can be catastrophic. A few millimetres in the wrong place can interrupt consciousness or stop breathing. This is also why brainstem death is used as a clinical definition of death.

1.4 Cranial Nerves

Most nerves reach the body via the spinal cord, but the twelve **cranial nerves** bypass that route and leave the CNS directly, mostly from the brainstem. They handle the close work of the head and neck, vision, eye movement, facial sensation and expression, taste, hearing, balance, swallowing, and, in the case of the vagus nerve, reach deep into the thorax and abdomen to regulate several visceral functions.

You do not need to memorise all twelve right now, but a few will keep coming up:

- The **optic nerve** (CN II) carries vision.
- The **oculomotor, trochlear, and abducens nerves** (CN III, IV, VI) move the eyes.
- The **trigeminal and facial nerves** (CN V, VII) carry facial sensation and drive facial expression.
- The **vestibulocochlear nerve** (CN VIII) carries hearing and balance.
- The **glossopharyngeal and vagus nerves** (CN IX, X) contribute to swallowing, speech, and parasympathetic regulation.

In the clinic, each cranial nerve can be tested individually at the bedside, which makes them a useful diagnostic window into the brainstem and connected structures.

1.5 Ventricles and Cerebrospinal Fluid

Inside the brain is a branching system of cavities called the **ventricles**: two lateral ventricles in the cerebral hemispheres, a third between them, and a fourth near the brainstem. They are filled with **cerebrospinal fluid (CSF)**, a clear fluid produced mostly by the *choroid plexus* that lines them.

It is tempting to think of CSF as just a liquid cushion, and that is certainly part of its job, the brain effectively floats in it, which dramatically reduces its effective weight and protects it against mechanical shock. But CSF also helps maintain a stable chemical environment around neurons and carries metabolic waste away from nervous tissue. It is manufactured in the ventricles, circulates out over the surface of the brain and spinal cord, and is eventually reabsorbed into the venous blood.

When CSF flow is obstructed, the ventricles swell under pressure, a condition called **hydrocephalus**, which we will meet again when we talk about ependymal cells.

1.6 The Spinal Cord

The spinal cord is a bundle of nervous tissue about the thickness of a finger that runs down inside the vertebral column. It does two things at once. First, it is a **highway**: sensory information travels upward to the brain, and motor commands travel downward to the body. Second, and less obviously, it is a **simple decision-making device** in its own right. Reflexes such as pulling your hand away from a hot surface are mediated by the spinal cord alone, without waiting for the brain to get involved. By the time you consciously register the pain, the hand has already moved.

The cord is segmented, and each segment gives rise to a pair of **spinal nerves** that serve a particular territory of the body. This segmental organisation is why spinal injuries at different levels produce such different patterns of loss: a cervical lesion affects the arms and everything below, while a lumbar lesion may spare the arms completely.

1.7 Peripheral Nerves and the Autonomic Nervous System

The peripheral nervous system extends from the CNS out into the rest of the body. Functionally, peripheral nerves are either **sensory (afferent)**, carrying information from the body back to the CNS, or **motor (efferent)**, carrying commands from the CNS out to muscles and glands.

Motor control itself splits into two worlds. The **somatic nervous system** governs the things you choose to do: moving an arm, speaking, walking. The **autonomic nervous system** governs the things you generally do not choose, heart rate, digestion, pupil size, sweating, blood pressure, and it does so through two opposing branches:

- The **sympathetic division** prepares the body for action: heart rate rises, pupils dilate, blood is shunted toward muscle. This is the “fight or flight” response.
- The **parasympathetic division** does the opposite: it slows the heart, promotes digestion, and generally encourages the body to conserve and repair. This is “rest and digest.”

This division matters far beyond the textbook. The balance between sympathetic and parasympathetic activity is the physiological signature of emotional arousal, stress, and relaxation, and it is where neurology and psychology start to meet in earnest: a frightened mind produces a sympathetically driven body, and a calm body feeds back into a calmer mind.

1.8 Summary

The nervous system is hierarchical and deeply integrated. The cerebrum handles the most elaborate cognition; the cerebellum refines motor output; the brainstem keeps the lights on; the spinal cord is both cable and reflex circuit; and peripheral nerves extend the system out into every organ and muscle in the body. With this map in hand, we can zoom in and ask how its individual building blocks, the neurons themselves, actually work.

2 Neuron Structure

Neurons are the cells that make the nervous system a nervous system. Every tissue in the body signals in one way or another, but neurons have specialised for one job to an extreme degree: they move information, quickly and reliably, over long distances. Everything else about their biology, their shape, their metabolism, their internal traffic system, is bent toward that single purpose.

The most striking consequence of this specialisation is that neurons are **polarised**. They are not blobs. They have a clearly defined input side and output side, and the whole architecture of the cell enforces one-way traffic. Understanding that shape is the first step toward understanding how neurons fire and communicate.

A neuron has three main regions, the **soma**, the **dendrites**, and the **axon**, plus a suite of internal machinery adapted to its unusual demands. We look at each in turn.

2.1 Soma, Dendrites, and the Axon

2.1.1 Soma (Cell Body)

The soma is the neuron's headquarters. It contains the nucleus and most of the cell's organelles, and it is where proteins are manufactured, energy is managed, and the general business of being a living cell is transacted. In a real sense the soma is the only part of the neuron that looks like a generic animal cell, everything else is specialised out.

Its most important computational job is **integration**. The soma, together with the axon hillock that we will meet in a moment, is where the neuron sums the inputs arriving on its dendrites and decides whether to fire. It is the place where "many inputs" becomes "one decision."

2.1.2 Dendrites

Dendrites are the branches that fan out from the soma like the root system of a tree, and they are the neuron's input antennae. A single neuron may have thousands of them, covered in small protrusions called *dendritic spines*, and each spine can host a synapse from another neuron. The sheer geometry of this arrangement means that one neuron can listen to thousands of others at once.

What arrives at a dendrite is not, at first, an electrical spike. It is a *chemical* signal, neurotransmitter released from the presynaptic cell, that is converted into a small, local change in membrane voltage called a **graded potential**. These graded potentials spread passively toward the soma, and they can either push the cell toward firing or pull it back from the brink. The dendritic tree is, in effect, the neuron's ear and its first stage of processing all at once.

2.1.3 Axon

The axon is the output. It is a single long projection that leaves the soma and carries electrical signals away to whatever the neuron talks to next, another neuron, a muscle cell, a gland. Axons can be astonishingly long: the motor neurons running from the spinal cord to the foot are over a metre in length in an adult, all in a single cell.

Three regions of the axon are worth naming explicitly:

- The **axon hillock** is the short region where the axon leaves the soma. It has an unusually high density of voltage-gated sodium channels, and it is here that action potentials are typically triggered. The hillock is, in effect, the neuron's "go / no-go" switch.
- The **axon shaft** conducts the action potential along its length.

- The **axon terminals** (or synaptic boutons) are the tiny swellings at the far end, where neurotransmitter is released onto the next cell.

Many axons are wrapped in **myelin**, a fatty insulating sheath produced by glial cells, which dramatically speeds up conduction. We will come back to this in the glial cells section, because it is a story worth telling properly.

2.2 Internal Machinery: Nucleus, Rough ER, and Golgi

Neurons are unusually demanding cells. They need a constant supply of ion channels, receptors, transporters, structural proteins, and neurotransmitter-related machinery, and they need to ship all of this out to parts of the cell that may be a metre away. The internal factory is built accordingly.

2.2.1 Nucleus

The nucleus holds the cell's DNA and controls gene expression. In neurons, that expression has to be finely tuned: the mix of receptors and channels a neuron produces is a large part of what determines its electrical personality, what it listens to, how it responds, and how it talks back.

2.2.2 Rough Endoplasmic Reticulum (Nissl Substance)

In a neuron, the rough endoplasmic reticulum is so abundant and so densely studded with ribosomes that it stains as a distinct pattern under the microscope. Historically this pattern was named **Nissl substance**. It is the neuron's protein factory: it synthesises the membrane proteins, ion channels, receptors, and secreted proteins that the cell needs in industrial quantities.

The amount of Nissl substance a neuron contains is a reasonable proxy for how metabolically active it is, and its dispersal (*chromatolysis*) is a classic sign of neuronal stress after injury.

2.2.3 Golgi Apparatus

The Golgi apparatus takes proteins fresh from the rough ER, modifies them, sorts them, and packages them into vesicles for delivery. In a neuron the logistics problem is particularly acute because proteins often need to be shipped far from the soma, and the Golgi is the shipping department at the head of the line.

2.3 Microtubules, Kinesin, and Dynein

Because a neuron's axon can be so long, ordinary diffusion is nowhere near fast enough to move molecules around inside it. Neurons solve this by building a system of molecular roads and motorised trucks.

2.3.1 Microtubules

The roads are **microtubules**: rigid, polar tubes of the cytoskeleton that run the length of the axon and into the dendrites. They provide mechanical support and, crucially, oriented tracks that motor proteins can walk along. The polarity matters, microtubules have distinct "plus" and "minus" ends, because it allows the cell to enforce direction on intracellular traffic.

2.3.2 Kinesin and Dynein

The trucks are motor proteins. Two families dominate axonal transport:

- **Kinesin** walks toward the plus end, which in an axon means away from the soma and out toward the terminals. This is *anterograde* transport, and it is how freshly made proteins, vesicles, and organelles are delivered to the synapse.
- **Dynein** walks the other way, toward the minus end and back toward the soma. This is *retrograde* transport, and it is how the cell recycles components, clears damaged cargo, and, sometimes, how signals from distant synapses make it back to the nucleus.

This bidirectional freight system is non-negotiable for neuronal life. A synapse at the end of a long axon is essentially a consumer that produces very little for itself; everything it uses has to be manufactured in the soma and delivered by kinesin, and everything it wants to get rid of has to be carried home by dynein. Disruption of axonal transport is implicated in several neurodegenerative diseases, which is hardly surprising once you realise how dependent the neuron is on this infrastructure.

2.4 Vesicles and SNARE Proteins

At the axon terminal, information has to cross from one cell to another, and for most neurons it does so chemically. Neurotransmitter is stored in advance inside small, membrane-bound **synaptic vesicles** that cluster just inside the terminal membrane, primed and ready to release.

Releasing a vesicle is not a casual event. It has to be fast, on the scale of fractions of a millisecond, and it has to be tightly coupled to the arrival of an action potential, or else the signal would be hopelessly noisy. The machinery that accomplishes this is a family of proteins called **SNAREs**.

SNAREs on the vesicle and SNAREs on the presynaptic membrane interlock like a zip, pulling the vesicle tight against the membrane and priming it for fusion. The final trigger is a rapid influx of calcium through voltage-gated channels that open when the action potential arrives. Calcium binds to a sensor protein, the SNARE complex completes fusion, and the contents of the vesicle are dumped into the **synaptic cleft**, the tiny gap between the two neurons, in a fraction of a millisecond.

This machinery is so central to nervous-system function that several of the most potent biological toxins known target it. Botulinum toxin and tetanus toxin, for example, both cleave specific SNARE proteins; the difference in their clinical effects comes down to which synapses they disable.

2.5 Neurotransmitters

A **neurotransmitter** is simply a chemical messenger released from one neuron that acts on receptors belonging to another. In principle the definition is simple; in practice the list of neurotransmitters is long and the consequences are enormous, because neurotransmitters are where neuroscience starts bleeding directly into psychology and psychiatry.

At the crudest level they fall into two groups:

- **Excitatory** neurotransmitters, the prototypical example being *glutamate*, push the post-synaptic cell toward firing.
- **Inhibitory** neurotransmitters, the prototypical example being *GABA*, pull it away from firing.

Most of your CNS signalling, by volume, is glutamate and GABA. Layered on top of this are the **neuromodulators**, chemicals like dopamine, serotonin, noradrenaline, and acetylcholine, which typically do not simply flip a downstream cell on or off but tune the way whole circuits respond to input. These are the systems most often implicated in mood, attention, reward, and arousal, and they are also the systems most commonly targeted by psychiatric drugs.

A rough working guide:

- **Dopamine**, movement, reward, motivation.
- **Serotonin**, mood, sleep, appetite.
- **Acetylcholine**, muscle activation at the neuromuscular junction; attention and memory in the brain.
- **Noradrenaline**, arousal and the alarm systems of the brain.

Whenever we later talk about a psychological phenomenon with a pharmacological correlate, depression treated with SSRIs, Parkinson's with L-DOPA, anxiety with benzodiazepines, we will be talking, in the end, about how a drug has tweaked one of these systems.

2.6 Summary

A neuron's shape is not incidental. The soma integrates, the dendrites listen, the axon transmits, and the terminals release chemistry onto the next cell. Inside the cell, a network of ER, Golgi, microtubules, and motor proteins keeps this long, demanding architecture supplied with everything it needs. At the synapse, vesicles wait ready and SNAREs stand by, so that when an action potential arrives, chemical communication can happen in a fraction of a millisecond. With this picture in place we are ready to ask the next question: how does a neuron actually fire?

3 Neuron Firing

So far we have been saying that neurons "send signals." It is time to be concrete about what that actually means. A neuron does not have tiny wires or its own battery. What it has is a membrane, some ions, and some very cleverly controlled channels through which those ions can move. Everything we call neural activity, every perception, every thought, every movement, is ultimately some pattern of ions crossing membranes.

This section builds the story up from the ground. We begin with the resting state: why an idle neuron has a voltage across its membrane at all. We then see how small, local disturbances (graded potentials) can add up to a threshold crossing, at which point a stereotyped, all-or-nothing **action potential** is born. Finally we follow that action potential down the axon to the synapse, where it triggers the chemical release we just described.

3.1 The Resting Membrane Potential

If you stick a very fine electrode inside an unstimulated neuron and another in the extracellular fluid, you find a voltage difference of roughly:

$$V_m \approx -70\,\text{mV}$$

The inside of the cell is negative relative to the outside. This is not an accident, and it is not "at rest" in the sense of nothing happening. Maintaining the resting potential is one of the most energy-hungry things a neuron ever does.

Three facts, taken together, explain it.

First, ions are distributed unequally across the membrane. Sodium (Na^+) is much more concentrated outside the cell than inside; potassium (K^+) is much more concentrated inside than outside; chloride (Cl^-) is more concentrated outside.

Second, the membrane is selectively permeable. At rest it is much more permeable to K^+ than to any other ion, because of leak channels that are always open.

Third, an active pump, the **Na^+/K^+ ATPase**, burns ATP to push three sodium ions out and pull two potassium ions in with every cycle. This both maintains the concentration gradients and contributes slightly to the negative interior.

Put these three facts together and the resting membrane potential emerges more or less inevitably. Because the membrane is dominated by potassium permeability, the cell's voltage sits close to the equilibrium potential for potassium.

3.1.1 The Nernst Equation

We can make this quantitative. For any single ion species, there is a particular membrane voltage at which the electrical force on the ion exactly cancels the diffusional force from its concentration gradient. This voltage is the ion's **equilibrium potential**, and it is given by the **Nernst equation**:

$$E_{ion} = \frac{RT}{zF} \ln\left(\frac{[ion]_{out}}{[ion]_{in}}\right)$$

Here R is the gas constant, T is temperature in kelvin, F is Faraday's constant, and z is the ion's charge. At body temperature (37°C), plugging in the constants and converting from the natural logarithm to the base-ten logarithm gives the convenient working form:

$$E_{ion} \approx \frac{61}{z} \log_{10}\left(\frac{[ion]_{out}}{[ion]_{in}}\right) \text{ mV}$$

3.1.2 Typical Equilibrium Potentials

For a mammalian neuron under physiological conditions:

- $E_K \approx -90$ mV
- $E_{Na} \approx +60$ mV
- $E_{Cl} \approx -65$ mV

Notice where -70 mV sits in that list. The resting potential is close to E_K because at rest, potassium is effectively the ion whose voice dominates. If a cell were instead to open a large number of sodium channels, the voltage would race toward $+60$ mV instead. This is exactly what happens during an action potential.

3.2 Graded Potentials

Before we reach the action potential, we need a smaller kind of signal: the **graded potential**. Graded potentials are the small, local changes in membrane voltage produced in dendrites and the soma by incoming synaptic input, and they are the raw material out of which the decision to fire is built.

Three properties define them. They are *graded*: their size depends on the size of the stimulus, not an all-or-nothing switch. They *decay* with distance: they are passive spreads of voltage, and they get weaker the further they travel. And they *summate*: if several graded potentials arrive at roughly the same time (temporal summation) or at nearby spots on the dendritic tree (spatial summation), their effects add together.

Two kinds are particularly important:

- **Excitatory postsynaptic potentials (EPSPs)** depolarise the cell, moving it closer to threshold.

- **Inhibitory postsynaptic potentials (IPSPs)** hyperpolarise the cell, moving it away from threshold.

The axon hillock tallies the EPSPs and IPSPs arriving at every moment, and when the net result is a depolarisation above threshold, an action potential is launched.

3.3 The Action Potential

The action potential is the headline event. It is a rapid, stereotyped, all-or-nothing spike in membrane voltage that propagates along the axon without decay, and it is how a neuron actually shouts across its own length.

3.3.1 Threshold

The action potential begins when the membrane at the axon hillock depolarises to about:

$$V_{\text{threshold}} \approx -55\,\text{mV}$$

At this voltage, voltage-gated sodium channels pop open in large numbers, and what follows is a runaway.

3.3.2 Phases of the Action Potential

It is best to think of the action potential as a short sequence of events, each dominated by a different set of channels.

1. Resting state (≈ -70 mV). Voltage-gated channels are closed. Leak channels and the Na^+/K^+ pump maintain the resting potential.

2. Depolarisation (up to $\approx +30$ mV). Voltage-gated Na^+ channels open in a positive-feedback avalanche: more sodium in means more depolarisation, which means more channels open. The voltage races upward toward E_{Na}.

3. Repolarisation. Two things happen, slightly delayed. Sodium channels *inactivate*, they physically plug themselves closed with an intrinsic gate, and voltage-gated potassium channels open. Potassium flows out and drags the voltage back down.

4. Hyperpolarisation (≈ -80 to -90 mV). Potassium channels are slow to shut, so K^+ efflux briefly overshoots the resting potential, pulling the cell closer to E_K.

5. Return to rest. Potassium channels close. The Na^+/K^+ pump continues its unending job of restoring the gradients, and the membrane settles back to -70 mV.

The entire spike, from threshold to recovery, takes only a few milliseconds. And because it is triggered by a self-reinforcing avalanche, it is all-or-nothing: a neuron either fires or it does not. There is no such thing as a partial action potential.

3.4 Neurotransmitter Release at the Synapse

When the action potential reaches the axon terminal, it finds yet another class of voltage-gated channels: **calcium channels**. These open, calcium floods in, and, as we saw in the previous section, that calcium is the trigger the SNARE machinery has been waiting for. Vesicles fuse with the presynaptic membrane and dump their neurotransmitter into the synaptic cleft, where it diffuses across and binds to receptors on the next cell.

The amount of neurotransmitter released depends on how much calcium enters, so the strength of the chemical signal is tightly linked to the arrival of the electrical one. In this way a digital, all-or-nothing spike is translated back into a graded chemical signal, which becomes a new graded potential in the next neuron, and the whole cycle repeats.

3.5 Clearing the Signal: Reuptake and Degradation

A signal is only useful if it can also be turned off. Neurotransmitter that is left to linger in the synaptic cleft blurs subsequent signals and, in the case of excitatory transmitters, can actually damage the postsynaptic cell (a phenomenon called *excitotoxicity*). There are three main ways the cleft is cleaned up:

- **Reuptake.** Specialised transporters on the presynaptic neuron (or on nearby glia) pump neurotransmitter back out of the cleft, often to be repackaged for reuse. This is the dominant mechanism for monoamines like serotonin and dopamine.
- **Enzymatic degradation.** At some synapses an enzyme in the cleft breaks the transmitter down. The classic example is *acetylcholinesterase*, which cleaves acetylcholine almost as soon as it is released.
- **Diffusion.** Simple loss of molecules out of the narrow cleft also contributes.

Many psychoactive drugs act here. Selective serotonin reuptake inhibitors (SSRIs) block the reuptake transporter for serotonin, raising its concentration in the cleft. Amphetamines interfere with dopamine reuptake and release. Nerve agents block acetylcholinesterase, producing catastrophic overstimulation. Whenever a drug is said to "increase" or "decrease" a neurotransmitter, it is almost always doing something very concrete and very local at the synapse.

3.6 Refractory Periods

After an action potential, the neuron is not instantly ready to fire another. For a brief interval it refuses absolutely, and for a slightly longer interval it only reluctantly agrees. These intervals are the refractory periods, and they are not bugs, they are what keep nervous-system signalling clean.

3.6.1 Absolute Refractory Period

Immediately after a spike, no new action potential can be generated at any price. This is because the voltage-gated sodium channels that just fired are *inactivated*: their inactivation gate is closed, and it will not reopen until the membrane has returned close to rest. No amount of stimulation will fire them. The absolute refractory period effectively sets an upper bound on how fast a neuron can fire.

3.6.2 Relative Refractory Period

During the hyperpolarisation that follows, sodium channels have recovered but the cell is further from threshold than usual. Firing is possible, but it takes a stronger-than-normal stimulus to get there.

Together, these two periods do two crucial jobs. They ensure that action potentials propagate **in one direction only**, the region just behind a travelling spike is refractory, so the spike cannot double back, and they cap the maximum firing rate, forcing the nervous system to encode intensity through *rate* and *pattern* rather than amplitude.

3.7 Summary

The electrical life of a neuron is a story of ions crossing a membrane. The resting potential is set mostly by potassium permeability. Small graded potentials from synaptic input summate at the axon hillock; if they push the membrane past threshold, a self-reinforcing avalanche of sodium entry produces an action potential, which propagates to the axon terminal. There, calcium influx triggers vesicle fusion, neurotransmitter is released, and the process starts again in the next cell. Refractory periods keep the signal unidirectional and bounded. The Nernst equation, which at first looks like a piece of physical chemistry, turns out to be quietly responsible for all of it.

4 Glial Cells

Read a textbook written fifty years ago and it might describe neurons as the stars of the nervous system and glial cells as their stagehands, quiet, supportive, uninteresting. That view is now dead. Modern neuroscience treats glia as active participants: they shape synapses, regulate ions, feed neurons, defend the brain, build the insulation that makes fast signalling possible, and fail or misbehave in many of the diseases we most dread. A surprisingly large portion of what we might loosely call "brain function" is not neuronal at all.

The word "neuroglia" literally means "nerve glue," which was a good guess in the nineteenth century and is a misleading one now. The cells are glue, and scaffolding, and plumbing, and police, and delivery driver, and insulator, and, increasingly, collaborator in information processing itself.

This section introduces the main types of glial cell, what each of them does, and where glial failure sits in neurological disease. The central theme is simple:

A neuron cannot do its job alone.

Every neuron in the body depends on glia to regulate its environment, recycle its transmitters, insulate its axon, and protect it from damage.

The main glial types, divided by location, are:

- **CNS**: astrocytes, oligodendrocytes, ependymal cells, microglia.
- **PNS**: Schwann cells, satellite cells.

4.1 Astrocytes and Satellite Cells

4.1.1 Astrocytes

Astrocytes are the workhorses of the CNS glial family. Under the microscope they are star-shaped (hence the name), with long processes radiating in every direction. Those processes contact neurons, synapses, blood vessels, and the surface of the brain, and that connectivity is the key to everything they do. An astrocyte sits at the intersection of almost everything important in nervous tissue.

Their jobs are numerous enough to be worth going through one by one.

1. Structural support. Astrocytes help hold the architecture of nervous tissue together. Their processes fill the spaces between neurons and give the CNS a kind of cellular scaffolding.

2. Ion buffering. Every time a neuron fires, potassium leaves the cell and accumulates briefly in the extracellular space. If this were allowed to build up, nearby neurons would become abnormally depolarised and fire uncontrollably. Astrocytes take up the excess potassium and redistribute it through their own network, a process called **spatial buffering**. Without it, neuronal activity would rapidly become chaotic.

3. Neurotransmitter recycling. Astrocytes are the main clean-up crew for glutamate, the principal excitatory transmitter of the brain. They pull glutamate out of the cleft through dedicated transporters, convert it to glutamine (which is inert and therefore safe), and ship the glutamine back to neurons to be re-converted into glutamate. This loop is called the **glutamate–glutamine cycle**, and it is the main defence against excitotoxicity.

4. Metabolic support. Neurons have extraordinary energy demands and almost no energy storage, so someone has to keep them fed. That someone is the astrocyte. Astrocytes take up glucose from blood vessels, store glycogen, and supply lactate and other substrates to neurons, an arrangement often called the *astrocyte–neuron lactate shuttle*.

5. The blood-brain barrier. The CNS is protected from the rest of the body by the **blood-brain barrier (BBB)**, a highly selective wall that only specific molecules can cross. The BBB is built chiefly by tight junctions between endothelial cells in brain capillaries, but astrocytic end-feet wrap around those vessels and are essential for inducing and maintaining the barrier's properties. Without astrocytes, the BBB does not form properly.

6. Modulation of synapses. The "tripartite synapse" concept treats every synapse as involving three cells, not two: the presynaptic neuron, the postsynaptic neuron, and an astrocytic process sitting alongside them. The astrocyte can detect neurotransmitter release, respond with its own calcium signals, and release *gliotransmitters* that feed back onto nearby neurons. How large a role this plays in moment-to-moment information processing is still an active research question, but the days of astrocytes as inert scaffolding are firmly over.

7. Response to injury. After any significant CNS insult, astrocytes become *reactive*. They enlarge, proliferate, change which genes they express, and in severe cases help form a dense **glial scar**. This response is double-edged: it walls off damage and limits inflammation, but the scar itself is a significant obstacle to the regeneration of neurons, which is one of the reasons CNS injuries heal so poorly.

4.1.2 Satellite Cells

Satellite cells are the PNS cousins of astrocytes. They form a close-fitting envelope around the cell bodies of sensory and autonomic neurons inside peripheral **ganglia**, such as the dorsal root ganglia. Their functions overlap substantially with those of astrocytes, structural support, control of the local chemical environment, regulation of ion concentrations, modulation of excitability, but they act on a much smaller, more localised scale.

Interest in satellite cells has grown sharply in recent years because of their role in chronic pain. Changes in satellite cell signalling are thought to contribute to the abnormal excitability of sensory neurons in persistent pain states, which makes them one of the more surprising targets on the emerging map of pain biology.

4.2 Oligodendrocytes

Oligodendrocytes are the myelinating glia of the CNS. Their name means "cells with few branches," a deliberate contrast with the exuberant astrocyte, and their central job is to wrap axons in **myelin**, the lipid-rich insulation that makes fast signalling possible.

A single oligodendrocyte can myelinate segments of several different axons at once, reaching out a process to each and wrapping it repeatedly. This is one of the most important differences between CNS and PNS myelin: a Schwann cell, as we will see, commits itself to a single internode on a single axon, while an oligodendrocyte hedges its bets.

4.2.1 What Oligodendrocytes Do

Their jobs are closely linked. They **produce myelin**, layer upon layer of plasma membrane wrapped tightly around the axon. They also **support axonal health** in ways that go beyond simple insulation, damage to oligodendrocytes tends to cause damage to the axons they serve, even when the axons themselves are initially untouched. And they **organise the axonal**

membrane into alternating **internodes** (myelinated) and **nodes of Ranvier** (bare gaps where voltage-gated sodium channels cluster).

The clustering of sodium channels at the nodes is what makes **saltatory conduction** possible. Instead of the action potential regenerating continuously at every point along the axon, it regenerates only at the nodes and effectively jumps from one to the next. The result is faster, more energy efficient, and more reliable.

4.2.2 Clinical Significance

The most important oligodendrocyte-related disease is **multiple sclerosis (MS)**, an autoimmune disease in which the immune system attacks CNS myelin. Axons that lose their myelin conduct slowly, unreliably, or not at all, and the resulting symptoms depend on where the damage falls: weakness, sensory disturbance, visual impairment, loss of coordination, fatigue. Some remyelination can occur, but repeated attacks cause accumulating and often permanent deficits.

MS is a powerful illustration of how central glia are to normal function. In MS, the neurons themselves are largely intact, and yet the patient is profoundly unwell, a reminder that the brain is not just its neurons.

4.3 Schwann Cells

Schwann cells are the PNS counterparts of oligodendrocytes, and they come in two flavours.

4.3.1 Myelinating Schwann Cells

A myelinating Schwann cell wraps itself repeatedly around a single segment of a single axon, producing one **internode** of myelin. One Schwann cell per internode, one axon at a time. Multiple Schwann cells are therefore needed along the length of any given PNS axon, separated by nodes of Ranvier just as in the CNS.

4.3.2 Non-myelinating Schwann Cells

Not every peripheral axon is myelinated. Small, slowly conducting axons (including many pain fibres and autonomic fibres) are instead embedded in non-myelinating Schwann cells, often multiple axons per cell, in structures called **Remak bundles**. These Schwann cells do not build concentric wrappings, but they still provide mechanical support, metabolic support, and protection.

4.3.3 Regeneration in the PNS

The single most striking difference between the PNS and the CNS is that peripheral nerves can regenerate and central nerves largely cannot. Schwann cells are a large part of why. After injury to a peripheral nerve, Schwann cells clear away the debris, proliferate, and form longitudinal tracks that actively guide regrowing axons back toward their targets, while secreting growth factors that promote regrowth. A cut peripheral nerve has a real chance of reconnecting; a cut spinal cord generally does not.

4.3.4 Clinical Significance

Peripheral demyelination and Schwann-cell dysfunction figure in several neuropathies, including **Guillain-Barré syndrome** (an autoimmune demyelinating attack, often following infection), **Charcot-Marie-Tooth disease** (a group of inherited neuropathies), and **diabetic neuropathy** (metabolic damage in long-standing diabetes). The common clinical picture is slowed nerve conduction, weakness, sensory loss, and reduced reflexes.

4.4 Myelin

Myelin is not a cell type in its own right, it is built by oligodendrocytes and Schwann cells, but it is so important, and so easy to misunderstand, that it deserves its own short discussion.

Physically, myelin is a multilayered sheath of plasma membrane wrapped tightly around an axon, squeezing out almost all of its cytoplasm so that what remains is essentially a stack of lipid bilayers. The stack is heavily lipid-rich (which is why myelinated tissue looks pale,*white matter*) and contains a characteristic set of structural proteins.

4.4.1 Why Myelin Speeds Up Conduction

There are two electrical tricks at work. First, wrapping an axon in many layers of membrane enormously **increases its resistance** to transverse current flow, which means less current leaks out across the membrane. Second, and equally importantly, it **decreases its capacitance**: a stack of membranes in series stores less charge than a single membrane, so less charge needs to be moved to change the voltage. Both effects mean that a given current spreads further and faster along the inside of the axon.

Instead of regenerating at every point, the action potential now jumps almost passively along the internodes, pausing at the nodes of Ranvier only long enough for sodium channels there to boost it before the next leap. The net result is dramatic: a well-myelinated axon can conduct at roughly 100 m/s, versus perhaps 1 m/s for an unmyelinated one of the same diameter.

4.4.2 Consequences of Demyelination

When myelin is stripped away, the electrical tricks disappear. Current leaks across the bare membrane, conduction slows, and action potentials may simply fail to propagate. This can cause severe neurological deficits even when the axons underneath are, for the time being, structurally intact, which is precisely what is so frightening about demyelinating diseases. Fix the myelin and the function can return; let the damage accumulate and the underlying axons may eventually die too.

4.5 Ependymal Cells

Ependymal cells form an epithelial-like lining along the walls of the ventricles and the central canal of the spinal cord. They are glial cells of a rather different flavour from astrocytes or oligodendrocytes: their job, essentially, is plumbing.

A specialised subset of ependymal cells, together with associated capillaries and connective tissue, forms the **choroid plexus**, the structure inside the ventricles that actively secretes cerebrospinal fluid. CSF is the fluid we met earlier: it cushions the brain, keeps its chemical environment stable, and removes waste products. Without ependymal cells, there is no CSF.

Beyond production, ependymal cells also help move CSF. Many of them have **cilia** on their apical surface, and the beating of these cilia drives circulation through the ventricular system. The ependymal lining also contributes to the regulation of exchange between CSF and brain tissue, though it is not as tight a barrier as the blood-brain barrier proper.

Clinically, anything that obstructs CSF flow or overwhelms its reabsorption can produce **hydrocephalus**, in which CSF accumulates inside the ventricles. The pressure rises, the ventricles balloon, and, especially in infants, whose skulls are still flexible, the head itself can visibly enlarge. Treatment usually involves surgically diverting CSF with a shunt.

4.6 Microglia

Microglia are the immune system of the CNS. They are unusual among glial cells in that they are not of ectodermal origin: they descend from the same mesodermal lineage as macrophages and other immune cells, and they migrate into the brain during early development. Once installed, they are essentially permanent residents.

In their "resting" state, microglia are not really resting at all. Their processes extend, retract, and extend again, continuously surveying the local tissue for anything out of place, pathogens, debris, abnormally folded proteins, damaged cells, inappropriate synaptic structures. When they find something, they respond.

4.6.1 What Microglia Do

Immune surveillance. Nothing fancy: they look, constantly, everywhere.

Phagocytosis. When activated, microglia engulf and digest dead cells, cellular debris, pathogens, and protein aggregates. They are the only professional phagocytes the CNS has.

Inflammatory signalling. Microglia release cytokines and chemokines to coordinate local inflammatory responses. In acute infection or injury this is protective. In chronic or excessive activation it is corrosive, and chronic microglial inflammation is increasingly implicated in neurodegenerative disease.

Synaptic pruning. During development, microglia physically remove unnecessary synapses, helping to refine neural circuits into their mature form. Remarkably, this pruning role continues into adulthood, and disruptions to microglial pruning are now suspected of contributing to several disorders of neurodevelopment.

4.6.2 Activation States

Microglial activation is not a simple switch. A resting microglial cell looks highly branched ("ramified"); an activated one retracts its branches and takes on a more amoeboid shape appropriate to its phagocytic role. But the states in between span a wide spectrum, and depending on context the same cell can be protective or destructive. The current picture is more one of *roles* than of discrete states.

4.6.3 Clinical Significance

Microglial dysfunction has been linked, in different ways, to a startling range of CNS diseases: Alzheimer's disease, Parkinson's disease, multiple sclerosis, stroke, traumatic brain injury, and several psychiatric conditions. In many of these, microglia contribute both to defence and to damage, they clean up after injury but can also sustain a chronic, smouldering inflammation that slowly erodes neurons. Figuring out how to tilt microglia toward their protective face and away from their destructive one is one of the hotter targets in current neurology.

4.7 Comparing CNS and PNS Glia

It is worth pausing to compare glial organisation in the two compartments.

In the CNS: astrocytes maintain homeostasis and support the BBB; oligodendrocytes myelinate; ependymal cells line the ventricles and make CSF; microglia provide immune defence.

In the PNS: Schwann cells myelinate (and non-myelinating Schwann cells envelop small axons); satellite cells support the cell bodies of ganglion neurons.

The parallels are not perfect, but the functional principles are the same, control the environment, insulate the axons, defend the tissue, and each system has specialised cells for each role.

4.8 Why Glia Matter

It is worth stating explicitly, because it is easy to forget:

- **Neurons need a tightly controlled environment**, and glia provide it, ions, neurotransmitters, extracellular composition, even energy substrates.
- **Fast signalling requires insulation**, and glia provide it through myelin.
- **The CNS needs protection**, and glia supply both its barrier (astrocytes and the BBB) and its standing immune defence (microglia).
- **Nervous tissue needs maintenance and repair**, and glia deliver most of it, including the PNS's remarkable capacity for regeneration.
- **Synaptic function is not purely neuronal.** Astrocytes and microglia actively shape synapses.

4.9 Summary

Glia are not the supporting cast of the nervous system, they are co-stars. Astrocytes and satellite cells regulate the environment in which neurons live. Oligodendrocytes and Schwann cells produce the myelin that makes fast signalling possible. Ependymal cells build and circulate the CSF. Microglia guard against injury and shape the wiring itself. Neuronal activity gets the headlines, but none of it would be possible without glial infrastructure. Any serious understanding of normal brain function, and of how the brain fails in disease, has to take the glial half of the story seriously.

5 The Frontal Lobe

We now have a working model of individual neurons, the signals they generate, and the glial tissue that supports them. It is time to zoom back out and look at one of the most interesting regions of the brain at a systems level, a region that has, more than any other, been identified with the person you feel yourself to be.

The **frontal lobe** is the largest of the cerebral lobes and sits just behind the forehead. It is responsible for voluntary movement, the planning and execution of actions, language production, and a cluster of higher cognitive and behavioural functions we collectively call *executive function*, judgment, decision-making, inhibition, personality. If the occipital lobe is where vision lives and the temporal lobe is where memory and hearing live, the frontal lobe is where thought is translated into action, and where action is held back in the service of longer-term goals.

It is also the region most implicated in the overlap between neurology and psychology, which makes it a natural place to end this set of notes. Damage to different parts of the frontal lobe produces characteristic and sometimes startling changes in behaviour and personality, changes that force us to confront how thoroughly the "mind" is rooted in very specific pieces of tissue.

5.1 Anatomical Landmarks

The frontal lobe is bounded by three anatomical landmarks you should be able to identify without hesitation:

- The **central sulcus** (Rolandic fissure) runs roughly down the top of the hemisphere and separates the frontal lobe from the parietal lobe. It is the most clinically important sulcus in the brain, because it divides the *motor* cortex in front from the *somatosensory* cortex behind.

- The **lateral sulcus** (Sylvian fissure) runs along the side of the hemisphere and separates the frontal lobe from the temporal lobe below.
- The **longitudinal fissure** runs down the midline and separates the two hemispheres from each other.

Within the frontal lobe, a few gyri (ridges) are worth knowing by name. The **precentral gyrus**, running immediately in front of the central sulcus, contains the primary motor cortex. The **superior, middle, and inferior frontal gyri**, running along the lateral surface, house progressively more abstract motor and cognitive areas as you move forward. Broca's area, for example, sits in part of the inferior frontal gyrus.

The rest of this section walks forward from the most concrete function, actually moving muscles, to the most abstract, deciding what kind of person you want to be.

5.2 Primary Motor Cortex (M1)

The **primary motor cortex**, traditionally labelled **M1** and corresponding to Brodmann area 4, occupies the precentral gyrus just in front of the central sulcus. It is the final cortical stage of voluntary movement: the place where the abstract intention to move becomes the concrete commands that drive muscles.

Its output leaves the cortex mainly via the **corticospinal tract** (for the body) and the **corticobulbar tract** (for the face and head), eventually reaching lower motor neurons in the brainstem and spinal cord that directly innervate muscles. Crucially, the corticospinal tract crosses to the opposite side of the body as it descends, so the left M1 controls the right side of the body and vice versa. This **contralateral** organisation is why a stroke in one hemisphere weakens the opposite side.

5.2.1 The Motor Homunculus

M1 is not a featureless sheet. It is organised *somatotopically*: different patches of cortex control different parts of the body, arranged in a roughly orderly map along the gyrus. This map is often drawn as a small distorted human figure, the **motor homunculus**, draped along the cortex.

Walking from medial to lateral along M1:

- The **medial** regions (down in the longitudinal fissure) control the lower limb.
- The **superior lateral** regions control the trunk and upper limb.
- The **lateral** regions control the face, lips, and tongue.

What makes the homunculus so striking visually is that it is wildly disproportionate. The hands and face occupy enormous stretches of cortex, while the trunk and legs get relatively little. The representation is proportional not to body size but to the fineness of motor control that part of the body demands. Hands and lips perform millisecond-level feats of precision; the small of your back does not.

5.2.2 Microanatomy

M1 contains exceptionally large pyramidal neurons called **Betz cells**, which project long axons into the corticospinal tract. These are among the biggest neurons in the entire CNS, an appropriate size for cells whose job is to reach all the way from the cortex to the spinal cord.

5.2.3 Clinical Picture

Damage to M1 (a stroke being the classic cause) produces contralateral weakness or paralysis along with loss of fine motor control. Because the lesion is above the level of the lower motor neuron, it produces so-called **upper motor neuron signs**: spasticity (excessive muscle tone), hyperreflexia (brisk reflexes), and an extensor plantar response (Babinski sign). The precise pattern of weakness depends on where within the homunculus the damage falls, a lesion over the "hand" region leaves the leg untouched, and vice versa.

5.3 Motor Association Cortex: Premotor and Supplementary Motor Areas

Immediately anterior to M1 lies the **motor association cortex** (Brodmann area 6), which is itself divided into two main regions: the **premotor cortex** on the lateral surface and the **supplementary motor area (SMA)** on the medial surface. These are the areas where movements are *planned* rather than executed. A useful way to think about them is that M1 decides how to move, and the motor association cortex decides what to move and in what order.

Their jobs include sequencing complex movements, coordinating the body so that posture and proximal muscles are prepared before the main action begins, and integrating sensory input into motor plans. There is a division of labour between them that is clinically worth remembering: the **supplementary motor area** is particularly important for *internally generated* and *bimanually coordinated* movements, the kind of movements you decide on for yourself, and the kind that require both hands to cooperate, while the **premotor cortex** is more involved in movements *cued by external stimuli*, like reaching for an object you just spotted.

Damage to these areas typically does not produce paralysis. It produces **apraxia**: a striking dissociation in which strength and coordination are intact, but the patient cannot perform learned purposeful movements on request. Asked to wave goodbye or mime brushing their teeth, they fumble. They still have the "how," in the low-level sense, but they have lost the "what", the motor plan. It is one of the cleanest demonstrations in neurology that planning and executing a movement live in different places.

5.4 Frontal Eye Field

Tucked into the posterior part of the middle frontal gyrus, and corresponding to Brodmann area 8, is the **frontal eye field (FEF)**. Its job is voluntary eye movement, in particular the rapid, ballistic eye movements called **saccades** that you use to jump your gaze from one point to another.

The FEF is how you direct your attention with your eyes. When you decide to look at something, the FEF is a major node in the network that makes it happen. It connects extensively to brainstem gaze centres and to the cranial nerve nuclei that actually drive the eye muscles. Stimulating one frontal eye field causes the eyes to move toward the opposite side; damaging it produces the opposite pattern, with the eyes drifting toward the side of the lesion because the healthy FEF on the other side is now unopposed. This "eyes toward the lesion" picture is a classic sign in the early hours of a frontal stroke.

5.5 Broca's Area

In the **inferior frontal gyrus** of the dominant hemisphere, which is almost always the left, even in most left-handers, lies **Broca's area**, corresponding to Brodmann areas 44 and 45. It is the part of the brain most closely associated with the *production* of language. It plans the motor sequences of speech, handles much of the grammar and sentence structure, and coordinates the muscles of the tongue, lips, larynx, and diaphragm that actually do the talking.

Damage to Broca's area produces **Broca's aphasia** (also called expressive or non-fluent aphasia). The clinical picture is characteristic and often heartbreaking: speech is slow, effortful, and telegraphic, content words survive, grammatical glue words drop out, and patients know, agonisingly, that what they are saying is not what they mean. Comprehension is relatively preserved, which is what distinguishes Broca's aphasia from the fluent but empty speech of Wernicke's aphasia, whose lesion lies further back in the temporal lobe.

Broca's area is also a reminder that cortical function is **lateralised**. The two hemispheres are not functional mirror images of one another. Language production sits overwhelmingly on the left for most people, while analogous regions on the right handle things like prosody and emotional tone.

5.6 Prefrontal Cortex

Moving still further forward, we arrive at the **prefrontal cortex (PFC)**: the portion of the frontal lobe that lies in front of the motor and premotor areas. This is the most recently evolved part of the frontal lobe and, in humans, proportionally the largest. It is also where the neurology of the frontal lobe bleeds most clearly into psychology.

The PFC does not produce movement directly. It produces the conditions under which movement makes sense. It holds goals in mind over time, compares possible courses of action, suppresses impulses that would be counterproductive, and integrates emotion, memory, and sensation into coherent behaviour. The whole constellation of abilities we call **executive function** lives here: working memory, planning, flexible switching between tasks, inhibition of prepotent responses, and the capacity to act in accordance with long-term goals rather than short-term reward.

5.6.1 Subdivisions

It is helpful to divide the PFC into three functional regions, each with its own characteristic contribution.

Dorsolateral prefrontal cortex (DLPFC). This is the region most associated with classical "cold" executive function: working memory, problem-solving, planning, and flexible control of attention. It is the part of the brain that lets you hold a phone number in mind while dialling, follow multi-step instructions, or mentally juggle the possibilities in a chess position. DLPFC dysfunction is implicated in attention deficits and is heavily involved in schizophrenia.

Ventromedial prefrontal cortex (VMPFC). This region is more concerned with "hot" cognition: emotion, risk, and reward. It integrates signals from the limbic system into decision-making, which is why damage here can leave formal reasoning intact while devastating the patient's ability to make real-life choices. Patients with VMPFC lesions can pass IQ tests and still ruin their lives through impulsive, poorly weighed decisions.

Orbitofrontal cortex (OFC). Sitting just above the orbits of the eyes, the OFC is closely involved in social behaviour, impulse control, and the moment-to-moment evaluation of rewards and punishments. It is especially important for recognising when something that used to be rewarding has stopped being so, and for holding back socially inappropriate behaviour.

5.6.2 Connections

The PFC is one of the most richly connected regions of the brain. It has strong reciprocal links with the limbic system (amygdala, hippocampus), with sensory and motor cortices, with the thalamus, and with the basal ganglia. This connectivity is what lets it perform its integrative role: executive function is, in a very literal sense, the business of pulling together information from everywhere else.

5.6.3 Phineas Gage and the Clinical Picture

No discussion of the prefrontal cortex is complete without **Phineas Gage**, a nineteenth-century American railway worker who survived an iron rod being driven through his frontal lobe in a blasting accident. Gage recovered physically and could walk, speak, and reason, but according to contemporary accounts his personality changed dramatically: the reliable, diligent foreman became impulsive, profane, and unable to hold down a job. The case is often overstated in popular retellings, but the core observation is real and, in its time, shocking: damage to a specific piece of brain tissue had produced a specific and durable change in who someone was.

The clinical syndromes seen after prefrontal damage map reasonably well onto the subdivisions above. DLPFC lesions tend to produce cognitive deficits: difficulty with working memory, planning, and task-switching, sometimes with apathy. VMPFC and OFC lesions tend to produce the more "Gage-like" picture: disinhibition, impulsivity, poor social judgment, emotional blunting, personality changes. In all cases the patient often lacks full insight into their own deficits, which is itself a kind of executive failure.

5.7 Summary

The frontal lobe is where the nervous system turns thought into action, and where action is disciplined in the service of goals. Its organisation reflects a rough gradient. The primary motor cortex at the back of the lobe executes movement; the premotor and supplementary motor areas plan and sequence it; the frontal eye field and Broca's area carry out specialised motor behaviours of gaze and speech; and the prefrontal cortex, sitting in front of all of these, supplies the executive control, judgment, and personality that make any of those actions meaningful. Damage at each level produces a characteristic and clinically revealing deficit, and together these deficits provide some of the most compelling evidence we have that the mind is, in the end, a function of specific pieces of brain tissue, which is both the starting point of neurology and the doorway through which psychology must pass.

6 The Parietal Lobe

Having mapped the frontal lobe, the cortex of action and intention, we now move backwards across the central sulcus into territory of a different character. The **parietal lobe** is the cortex of *sensation* and *space*: where the body's tactile world is registered, where the position of every limb is tracked moment by moment, and where the brain assembles its sense of where things are, including where you yourself are within the world.

Where the frontal lobe pushes outward, sending commands to muscles, the parietal lobe pulls inward, taking the river of information arriving from the skin, joints, muscles, and viscera and turning it into a coherent picture of the body and its surroundings. It is also the place where touch meets vision, where felt position meets seen position, and where the multiple sensory streams are bound into the unified spatial scene that we treat, naively, as simply "the world."

6.1 Anatomical Landmarks

The parietal lobe sits behind the frontal lobe and above the temporal lobe, occupying roughly the upper-rear quadrant of each hemisphere. Its boundaries are:

- Anteriorly, the **central sulcus** separates it from the frontal lobe. The first ridge behind the sulcus, the **postcentral gyrus**, mirrors the precentral gyrus in front and contains the primary somatosensory cortex.
- Posteriorly, the **parieto-occipital sulcus**, visible most clearly on the medial surface, marks the boundary with the occipital lobe.

- Inferiorly, the **lateral sulcus** (Sylvian fissure) separates it from the temporal lobe.
- Medially, the **longitudinal fissure** divides it from its counterpart on the other side.

Within the parietal lobe, the most important subdivision is made by the **intraparietal sulcus**, which runs roughly horizontally and divides the surface into a **superior parietal lobule** above and an **inferior parietal lobule** below. The inferior parietal lobule is itself made up of two named gyri: the **supramarginal gyrus**, which curls around the end of the lateral sulcus, and the **angular gyrus**, which sits behind it and curls around the end of the superior temporal sulcus. These are landmarks worth keeping in mind, as several of the more striking parietal syndromes localise to them.

Functionally, the parietal lobe walks from the most concrete to the most abstract as you move backwards. The postcentral gyrus, just behind the central sulcus, registers raw sensation; the cortex behind it interprets and integrates that sensation; and the cortex further back still uses sensation to construct space, attention, and bodily self-awareness.

6.2 Primary Somatosensory Cortex (S1)

The **primary somatosensory cortex**, traditionally labelled **S1** and corresponding to **Brodmann areas 3, 1, and 2**, occupies the postcentral gyrus immediately behind the central sulcus. It is the cortical destination of the body's sensory traffic, the place where information about touch, pressure, vibration, joint position, and pain finally becomes conscious sensation.

The information arriving at S1 has come a long way. **Discriminative touch, vibration, and proprioception** ascend through the **dorsal column–medial lemniscus pathway**; **pain and temperature** ascend through the **spinothalamic tract**. Both pathways relay through the **thalamus**, specifically the **ventral posterolateral (VPL) nucleus** for the body and the **ventral posteromedial (VPM) nucleus** for the face, before fanning out to S1. As with the motor system, the projection is **contralateral**: the right S1 represents the left side of the body and vice versa.

6.2.1 The Sensory Homunculus

S1 is somatotopically organised in just the same way M1 is, and the resulting map is called the **sensory homunculus**. Walking from medial to lateral along the postcentral gyrus:

- The **medial** regions, tucked into the longitudinal fissure, represent the lower limb and genitals.
- The **superior lateral** regions represent the trunk and upper limb.
- The **lateral** regions represent the face, lips, and tongue.

Like its motor counterpart, the sensory homunculus is grotesquely disproportionate. The lips, tongue, fingertips, and face occupy enormous swathes of cortex; the back, trunk, and legs make do with much less. The principle is the same as in M1: cortical real estate scales with the *density of receptors* and the *fineness of discrimination* demanded of that body part, not with its physical size. The fingertips can distinguish points a millimetre or two apart, and they require the cortex to do it.

6.2.2 Microanatomy

The three Brodmann subdivisions of S1 are not redundant. **Area 3** is the most anterior strip, sitting in the wall of the central sulcus, and it is the principal recipient of thalamic input. It is itself split into **3a**, which receives proprioceptive information from muscles and joints, and

3b, which receives cutaneous information from the skin. Behind these, **area 1** processes mainly cutaneous input as well, and **area 2** integrates cutaneous and proprioceptive signals to support more complex perceptions such as size, shape, and texture. There is, in other words, an early stage of integration happening already within S1 itself.

6.2.3 Clinical Picture

Damage to S1, again most often from a stroke, produces **contralateral sensory loss**, with the most prominent losses being in the modalities S1 is best at: fine discriminative touch, vibration, proprioception, and two-point discrimination. Crude awareness of touch and pain often survives, because thalamic and other subcortical structures contribute their own coarse processing, but the patient loses the ability to feel *where* they are touched precisely, to recognise objects by feel, or to know without looking exactly where their hand is in space. The pattern of loss respects the homunculus: a small lesion in the "hand" region of S1 leaves the leg untouched.

6.3 Somatosensory Association Cortex

Behind S1, on the **superior parietal lobule**, lies the **somatosensory association cortex**, corresponding to **Brodmann areas 5 and 7**. Where S1 registers raw sensation, the association cortex *interprets* it. It takes the disjointed feel of edges, textures, weights, and contours arriving from S1 and assembles them into the recognition of objects, the sense of one's body shape, and the awareness of where each limb is in space.

A useful way to think about it is that S1 tells you *that* something is touching you and *where*, while the association cortex tells you *what it is*. Closing your eyes and identifying a key in your pocket, distinguishing a coin from a button by feel, sensing the heft of a tool well enough to use it, all of this is somatosensory association cortex at work. It also stores tactile memories, so that the feel of familiar objects becomes immediately recognisable rather than having to be reconstructed from scratch each time.

Damage to this region produces a striking deficit known as **tactile agnosia**, or, when restricted to a hand, **astereognosis**. Strength is intact, primary sensation is intact, and the patient can describe the temperature, texture, and shape of an object placed in the hand, but cannot identify what the object is. The raw materials of perception are present; the act of recognition has been severed from them. As with the motor association cortex's apraxia, this is a clean dissociation between low-level capability and the higher-level integration that makes it useful.

6.4 Posterior Parietal Cortex

Further back still, occupying much of the rest of the parietal lobe and especially the inferior parietal lobule, lies the **posterior parietal cortex (PPC)**. This is the most abstract of the three regions and the one that bleeds most clearly into cognition. Its job is to construct **space** and **attention**: to integrate visual, somatosensory, and vestibular information into a unified sense of where things are, where the body is among them, and where attention should be directed next.

Practically, the PPC is what allows you to reach accurately for a cup without staring at your hand, to navigate a doorway without bumping the frame, to keep track of a moving object while turning your head, and to feel that the room is stable even though your retinal image lurches with every saccade. It is also a major node in the brain's **attention network**, working with the frontal eye field and other regions to allocate attention to relevant parts of the environment.

The PPC is, like the frontal lobe, strongly **lateralised**, and the syndromes produced by damage on each side differ instructively.

6.4.1 Hemispatial Neglect

Damage to the **right (non-dominant) posterior parietal cortex** produces **hemispatial neglect**, one of the most arresting syndromes in neurology. Patients fail to attend to the left half of space, not because they are blind on the left, their visual pathways are intact, but because the part of the brain that would direct attention there has been silenced. They eat from only the right side of the plate, shave only the right side of the face, copy only the right half of a clock, and may deny that the left arm, lying inert beside them, is theirs at all. The world has been cut in half not at the level of perception but at the level of attention.

The asymmetry, that right PPC damage neglects the left while left PPC damage rarely produces a comparable right-sided neglect, is usually explained by the right hemisphere's more global role in spatial attention: the left attends mainly to the right side of space, but the right attends to both sides, so losing the right is catastrophic in a way that losing the left is not.

6.4.2 Gerstmann Syndrome

Damage to the **left (dominant) inferior parietal lobule**, particularly around the **angular gyrus**, can produce **Gerstmann syndrome**: a tetrad of **acalculia** (inability to do simple arithmetic), **agraphia** (inability to write, with intact speech and comprehension), **finger agnosia** (inability to identify or distinguish one's own fingers), and **left–right disorientation**. The syndrome is rarely complete, and its precise localisation has been debated, but the cluster is clinically real and points to the role of the dominant inferior parietal lobule in the kinds of abstract symbolic and bodily representations that arithmetic, writing, and the body schema all depend on.

6.4.3 Optic Ataxia and Balint's Syndrome

Lesions of the superior parietal lobule and the surrounding PPC can produce **optic ataxia**, a striking inability to reach accurately for visually presented objects despite intact vision and intact motor strength. The eyes see the cup, the arm can move, but the visuospatial transformation that converts "the cup is there" into "my hand should go here" has been lost. Optic ataxia is the central feature of **Balint's syndrome** when bilateral PPC damage is present, alongside **simultanagnosia** (inability to perceive more than one object at a time) and **ocular apraxia** (inability to direct gaze voluntarily). Together these deficits offer a glimpse of how thoroughly the experience of a coherent visual scene depends on parietal machinery.

6.5 Summary

The parietal lobe is where the body's sensory traffic becomes conscious sensation, where sensation becomes recognition, and where recognition is woven together with vision and movement into the felt sense of a body in space. The primary somatosensory cortex registers touch, proprioception, pain, and temperature in a contralateral, somatotopically organised map; the somatosensory association cortex turns those raw sensations into recognised objects and a coherent body schema; and the posterior parietal cortex constructs space, attention, and the visuospatial frames within which action becomes possible. Damage at each level produces a characteristic deficit, from contralateral numbness, through tactile agnosia, to the more dramatic syndromes of neglect, Gerstmann's tetrad, and Balint's. Taken together, the parietal lobe shows that even something as immediate as "what the world feels like" is the product of layered cortical processing, and that the mind's sense of being embodied in a place is not given but built.

7 The Temporal Lobe

Dropping below the lateral sulcus, we leave the cortex of touch and space and enter the cortex of *sound*, *smell*, and *meaning*. The **temporal lobe** is where vibrations of air become recognised speech and music, where the volatile chemistry of the world becomes the pull of a familiar smell, and where, in its language regions, the heard word becomes the understood word.

If the frontal lobe turns intention into action and the parietal lobe turns sensation into space, the temporal lobe turns sensation into *recognition* and *meaning*. It is where the auditory and olfactory streams arrive in cortex, where they are interpreted, and where the interpretation of language as comprehensible content first becomes possible. It is also, along the way, the cortex most tightly bound to the limbic system, which is part of why sound and smell are so good at evoking memory and emotion.

7.1 Anatomical Landmarks

The temporal lobe sits below the lateral sulcus and forwards of the occipital lobe, occupying roughly the lower-side of each hemisphere. Its principal external landmarks are three roughly parallel gyri running along its lateral surface:

- The **superior temporal gyrus**, immediately below the lateral sulcus, contains the auditory cortices and the receptive language area in the dominant hemisphere.
- The **middle temporal gyrus**, beneath it, contributes to higher-order auditory and visual processing, including aspects of object and motion recognition.
- The **inferior temporal gyrus**, lower still, contributes to high-level visual recognition, particularly of faces and objects.

Two further landmarks matter. Tucked into the floor of the lateral sulcus on the upper surface of the superior temporal gyrus lie **Heschl's gyri**, also called the **transverse temporal gyri**, which house the primary auditory cortex itself. On the medial surface, the temporal lobe curls inward to form the **uncus** and the **parahippocampal gyrus**, structures continuous with the limbic system; the primary and association olfactory cortices live here, alongside the hippocampus that we will meet in a later section on memory.

As with the other lobes, the temporal lobe walks from concrete to abstract: primary sensory cortex first, association cortex behind it, and language and memory regions further back and inward.

7.2 Primary Auditory Cortex (A1)

The **primary auditory cortex**, traditionally labelled **A1** and corresponding to **Brodmann areas 41 and 42**, occupies Heschl's gyri on the upper surface of the superior temporal gyrus, mostly hidden inside the lateral sulcus. It is the cortical destination of the auditory pathway, where pressure waves in the ear finally become the conscious experience of sound.

The pathway that delivers sound to A1 is one of the longest and most heavily processed in the brain. Hair cells in the **cochlea** convert mechanical vibrations into neural signals carried by the **cochlear division of the vestibulocochlear nerve (CN VIII)**. From there, signals pass through the **cochlear nuclei** of the brainstem, the **superior olivary complex**, the **lateral lemniscus**, the **inferior colliculus** of the midbrain, and the **medial geniculate nucleus (MGN)** of the thalamus before finally fanning out to A1. At several points along this chain, information crosses the midline, with the result that each A1 receives input from *both* ears. This bilateral organisation is why a unilateral lesion of A1 does not produce deafness in either ear; it produces only a subtle deficit in tasks like sound localisation and the perception of complex auditory scenes.

7.2.1 Tonotopic Organisation

A1 is not a featureless sheet either. It is organised **tonotopically**: different patches of cortex respond to different frequencies, arranged in an orderly map that mirrors the mechanical tuning of the cochlea. Roughly, **high frequencies** are represented at one end of A1 and **low frequencies** at the other, with a smooth gradient between. This is the auditory analogue of the somatotopic organisation of S1 and the retinotopic organisation of the visual cortex: a peripheral sensory surface mapped, in orderly fashion, onto the cortex.

7.2.2 Clinical Picture

Because of the bilateral pathway, unilateral A1 damage rarely causes obvious hearing loss. What it does produce are problems with *spatial hearing* (working out where a sound came from) and with the perception of complex auditory scenes, particularly on the side opposite the lesion. **Cortical deafness**, in which hearing is essentially absent despite an intact ear and intact brainstem auditory pathway, requires the rare misfortune of **bilateral** A1 damage.

7.3 Auditory Association Cortex

Surrounding A1 on the superior temporal gyrus, and corresponding to most of **Brodmann area 22**, lies the **auditory association cortex**. Where A1 registers raw sound, the association cortex *interprets* it: it recognises the sound as speech, music, a familiar voice, a doorbell, or a piece of meaningless noise. It is also the region that gives auditory experience much of its temporal structure, the perception of rhythm, melody, and the patterning of language at the level of pure sound.

Damage to the auditory association cortex produces **auditory agnosia**, a deficit analogous to the tactile agnosia of the parietal lobe. The patient hears: pure-tone audiometry is intact, A1 is doing its job, sound is consciously perceived, but recognition has been severed from perception. Specific subtypes are clinically recognised. **Pure word deafness** leaves the patient unable to understand spoken language while still able to hear, read, write, and speak normally; non-speech sounds are recognised, but speech is reduced to noise. **Amusia** leaves the patient unable to recognise melodies that were once familiar, or, in some cases, unable to perceive music as music at all. As with other association-cortex syndromes, what has been lost is not the input but the act of making the input meaningful.

7.4 Primary Olfactory Cortex

Smell is the odd one out among the senses, and the temporal lobe is where its peculiarity is on display. The **primary olfactory cortex** sits on the medial surface of the temporal lobe, primarily on the **uncus** and adjacent piriform cortex, and includes parts of the **piriform cortex**, periamygdaloid cortex, and amygdala. Its inputs come from the **olfactory bulb** via the **olfactory tract (CN I)**, the only cranial nerve that connects directly to the cerebrum without first passing through the brainstem.

What makes olfaction unusual is that, alone among the major senses, it does not relay through the thalamus before reaching cortex. Olfactory receptor neurons in the nasal epithelium project to the olfactory bulb, the bulb projects to the primary olfactory cortex directly, and only *after* this initial cortical processing does smell information reach the thalamus and the orbitofrontal cortex for further integration. This shortcut is part of why smell has such an immediate, unmediated quality, and why it is so tightly bound to emotion and memory: the primary olfactory cortex is essentially continuous with the limbic system.

7.4.1 Clinical Picture

Damage to the primary olfactory cortex, or to the olfactory tract that feeds it, produces **anosmia**, the loss of the sense of smell, often noticed first as a loss of flavour in food (since flavour is largely olfactory). Irritative lesions of this region, especially around the uncus, can produce the opposite picture: brief, unpleasant olfactory hallucinations known as **uncinate fits**, classically a vivid smell of burning rubber that heralds a temporal lobe seizure. The smell is hallucinated, but the underlying electrical disturbance is unmistakable.

7.5 Olfactory Association Cortex

Adjacent to and continuous with the primary olfactory cortex, the **olfactory association cortex** occupies the **entorhinal cortex** (Brodmann area 28) and parts of the parahippocampal gyrus. Its job is to take the raw olfactory signal arriving at primary cortex and turn it into a recognisable, named, emotionally weighted experience: this smell is coffee, that one is petrol, this one is the kitchen of someone you have not seen in twenty years.

The olfactory association cortex is the most direct anatomical demonstration of why smell is so tied to memory and feeling. Through the entorhinal cortex it has powerful, reciprocal connections with the **hippocampus** (the seat of declarative memory) and the **amygdala** (a key node in emotional processing). A familiar smell can summon a specific memory with a vividness that other senses rarely match, partly because the wiring that delivers it is, in a literal sense, two synapses away from the structures that store autobiographical memory and assign emotional valence.

Damage here produces the recognition counterpart to anosmia: smells are still consciously detected but cannot be identified or attached to memory, an olfactory analogue of the agnosias seen elsewhere. Entorhinal cortex is also one of the very first regions to degenerate in **Alzheimer's disease**, which is part of why a degraded sense of smell can be one of the earliest non-cognitive signs of the illness.

7.6 Wernicke's Area

In the **posterior part of the superior temporal gyrus** of the dominant hemisphere, almost always the left, lies **Wernicke's area**, traditionally identified with the posterior portion of **Brodmann area 22**. It is the part of the brain most closely associated with the *comprehension* of language. If Broca's area, in the inferior frontal gyrus, is where the motor plans for speech are assembled, Wernicke's area is where heard and read words become understood meaning.

Anatomically, Wernicke's area sits where it does for a reason: it is right next door to the auditory association cortex, perfectly placed to interpret the patterned sounds of speech, and it is connected to Broca's area by a long-range white-matter tract called the **arcuate fasciculus**. The arcuate fasciculus is what allows the comprehended word to be turned, almost without effort, into the spoken word: it ferries information from the back of the language network to the front.

7.6.1 Wernicke's Aphasia

Damage to Wernicke's area produces **Wernicke's aphasia**, also called **receptive** or **fluent aphasia**, and the clinical picture contrasts instructively with Broca's. Speech is *fluent*: words come out at a normal rate, with normal grammatical scaffolding and normal prosody. But the content is degraded. Patients substitute words (*paraphasias*), invent words (*neologisms*), and at the extreme produce a fluent stream of well-formed-sounding but largely meaningless language sometimes called *word salad*. Comprehension is also impaired, often severely, and, crucially, patients frequently lack insight into their own deficit. Whereas a Broca's patient agonisingly knows that what they are saying is not what they mean, a Wernicke's patient often does not.

The contrast is one of the cleanest demonstrations in neurology that language production and language comprehension are anatomically separable, and that fluency and meaning are not the same thing.

7.6.2 Conduction Aphasia

Damage that spares both Wernicke's and Broca's areas but cuts the **arcuate fasciculus** between them produces **conduction aphasia**: comprehension is largely intact, speech is largely fluent, but the patient is strikingly unable to *repeat* what has just been said. The information cannot make its way from the comprehending cortex of the temporal lobe to the producing cortex of the frontal lobe, even though both are functioning. It is the strongest clinical evidence we have that the language network is not just a set of regions but a set of regions linked by specific tracts.

7.7 Summary

The temporal lobe is where the brain turns sound into meaning, smell into memory, and heard speech into understood language. The primary auditory cortex on Heschl's gyri receives the bilateral output of the auditory pathway and arranges it tonotopically; the auditory association cortex of the superior temporal gyrus turns that raw sound into recognisable speech, music, and familiar voices. The primary olfactory cortex on the uncus, alone among the senses, receives its input directly without a thalamic relay, and the olfactory association cortex of the entorhinal region binds smell to memory and emotion through its links with hippocampus and amygdala. Finally, Wernicke's area in the dominant superior temporal gyrus stands at the centre of language comprehension, paired with Broca's area through the arcuate fasciculus to form the language network. Damage at each of these levels produces a characteristic deficit, from cortical deafness and anosmia, through auditory and olfactory agnosias, to the contrasting aphasias of Wernicke and the disconnection picture of conduction aphasia. Together, the temporal lobe is where the mind acquires much of its capacity for meaning.